

CA-Software Assignment - 1 :

Design Pattern Explanation - Prepare a one-page summary explaining the MVC (Model-View-Controller) design pattern & its two variants. Use diagrams to illustrate their structure & briefly discuss when each variant might be more appropriate to use than the others.

MVC is a software design pattern that separates an application into three interconnected components:

- Model: Manages the data, logic & rules of the application.
- View: Displays data (UI) & sends user commands to the controller.
- Controller: Handles user input, interacts with the model, & updates the view.

Standard MVC Structure Diagram:

[User] → [Controller] → [Model] → View

Variants of MVC :

1. Model-View-Presenter (MVP)

- Model: Same as MVC
- View: More passive; only displays what the Presenter tells it.

- Presenter: Acts as a Mediator between view & Model.
contains the presentation logic.

MVP Diagram:

[User] $\rightarrow$ [view] $\rightarrow$ [presenter] $\rightarrow$ Model.

When to Use:

- In desktop or mobile applications where separation of concerns in UI is needed.
- Useful when testing is a priority,

2. Model-View-ViewModel (MVVM)

- Model: Data & business logic
- View: UI layers, binds to properties of the ViewModel.
- ViewModel: Exposes data & commands to the View using data binding.

MVVM Diagram:

[view] $\leftrightarrow$ [viewModel] $\leftrightarrow$ [model]

When to Use:

- In platforms like Angular, React, or WPF with strong support for data binding.
- Useful in reactive UI systems.

Model comparison & Usage:

pattern	Best Use Case
MVC	Web applications like Django or Ruby on Rails
MVP	Apps with complex UI logic & testing needs
MVVM	frameworks with binding (WPF, Angular, React)